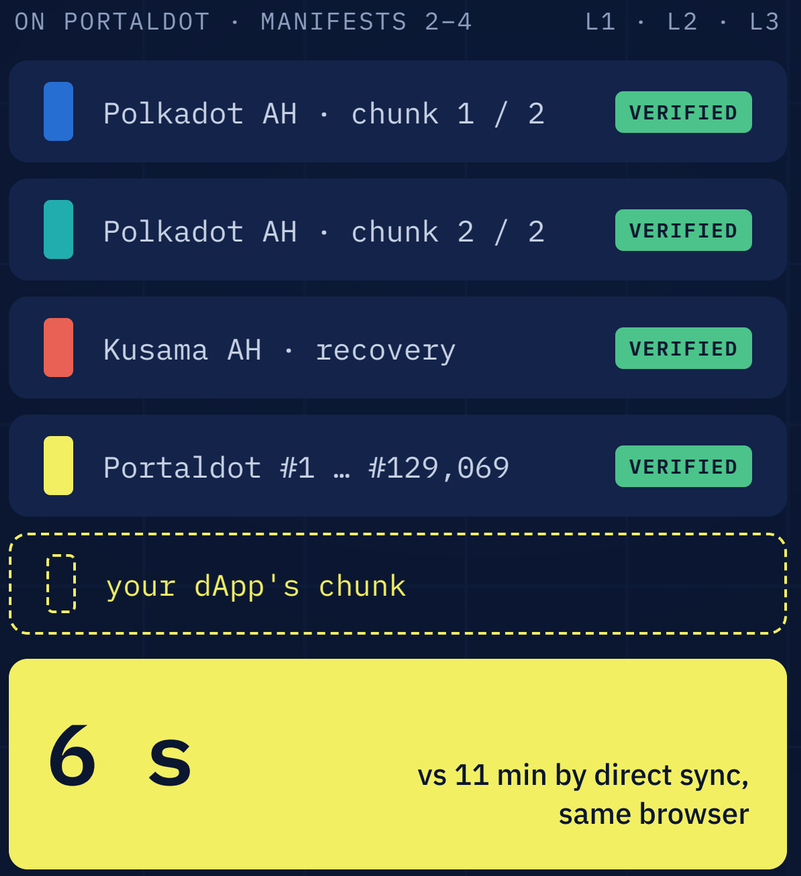
potluck

PORTALDOT HACKER HOUSE 2026 · MODULAR CLOUD & MIDDLEWARE

Everyone brings a chunk. Every dApp gets *the history.*

On-chain history for any dApp, loaded in seconds in the browser with no servers. Stored and guaranteed by Portaldot, verified against the chain it came from.



HISTORY IS THE MISSING API

dApps need history. Nodes only answer *what is true now.*

01 Every dApp runs or rents an indexer server to show transfers, holders and timelines.

02 Users must trust that server's database. Nothing checks it against the chain.

03 When the server goes away, the history goes with it.

2–4 days

to sync Polkadot Asset Hub's Revive window (~10.65 M blocks) naively from a browser over public RPC.

22.05 M
Polkadot Asset Hub blocks, ~3.0 events each

22.78 M
Kusama Asset Hub blocks, ~4.7 events each

Measured 2026-10-20 on public archive nodes

POTLUCK

Index once, store it on Portaldot, *verify it everywhere.*

01 · PUBLISH

Anyone indexes

A publisher runs a declarative JSON spec against a source chain's archive RPC, now and then. Output: canonical NDJSON chunks, identical on every machine.

Who: anyone with the CLI and a little POT. Not an always-on server.

02 · STORE

Portaldot keeps it

Chunks go into TransactionStorage, where validators must keep proving they hold the data. A manifest is registered in PotluckRegistry on pallet-revive.

Who: Portaldot validators. Paid once, read by anyone for free.

03 · VERIFY

The browser checks

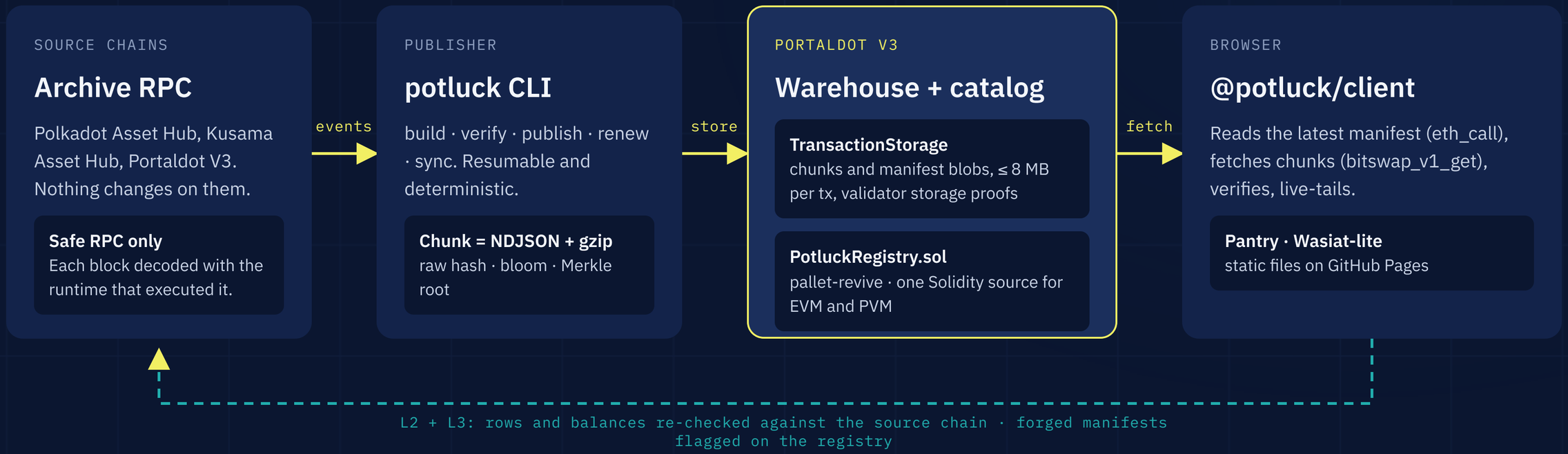
The dApp fetches only the chunks it needs over the node's bitswap_v1_get, checks them against the source chain, then live-tails new blocks.

Who: the user's own browser. No API key, no backend.

Snapshots are only an accelerator. With no manifest, the SDK syncs straight from the source chain: slower, never blocked.

ARCHITECTURE

Four parts, *zero servers.*



THREE LEVELS, CHECKED IN THE BROWSER

Trust the source chain, *not the publisher.*

L1 **Integrity**

Digest, raw hash, canonical encoding, bloom. Catches corrupted or swapped chunks.

L2 **Row truth**

Each row re-derived from its block's events on the source chain. Catches invented or altered rows.

L3 **Reconciliation**

Balance before + every row's effect = balance after, for every account and window. Catches hidden rows, even with every hash forged.

ATTACK DEMO · PANTRY

A forged chunk hides one transfer and recomputes every hash.

L1	hashes, encoding	PASSES
L2	rows that remain	PASSES
L3	903 / 904 accounts	CAUGHT

The one failing account's difference is **exactly the hidden amount.** The manifest is flagged on-chain and clients skip it.

Balance reconciliation measured exact on 319 / 319 sampled Polkadot AH accounts (spike S4)

WHY PORTALDOT IS ESSENTIAL

The only warehouse that is permissionless, guaranteed *and browser-readable.*

WHERE HISTORY COULD LIVE	STORE WITHOUT PERMISSION	WHO GUARANTEES THE BYTES	READ STRAIGHT FROM A BROWSER
Portaldot TransactionStorage	**Yes.** Fee only, ≈ 100 tPOT per MB, ≤ 8 MB per tx	**Validators**, who must prove they still hold it every block	**Yes.** bitswap_v1_get on any node, no gateway
Polkadot Bulletin Chain	**No.** Per-account authorization quota	Validators	Only for authorized uploaders' data
IPFS	Yes	**Whoever keeps pinning**: a server	Through a gateway or a local node
An indexer's database	Operator only	**The operator**, while they pay	Through their API, trusted as-is

The manifest registry lives on **pallet-revive**, read with a plain eth_call. This matches Portaldot's own pitch as a Layer0 that moves data between protocols.

Retention 100,800 blocks (~7 days); anyone can renew for ≈ 0.012 tPOT per blob

WASM/EVM COMPATIBILITY · MODULAR · MULTI-CHAIN

One product, *every V3 primitive.*

TransactionStorage

- Chunks, specs and manifests stored on-chain; fees measured on testnet
- Blobs fetched by CID with bitswap_v1_get from the public node: 2.5 MB in ≈ 4 s

pallet-revive, both VMs

- PotluckRegistry.sol: one source, solc → EVM (deployed) and resolc → PVM (dry run passes)
- Identical ABI, so the client does not care which VM serves it

Two-VM history for Portaldot

- Native transfers + Revive ERC-20/721 logs in one index
- SS58 ↔ H160 from the ReviveApi runtime API; 14 ↔ 18 decimals handled

Multi-chain sources

- Polkadot AH, Kusama AH and Portaldot V3 through one spec format
- Every block decoded with the runtime that executed it (≥ 35 Polkadot AH runtimes)

Adaptive sync. Direct when cheap (Portaldot today), verified snapshots when not. When Portaldot gets busy, its dApps switch over with no code changes.

MEASURED ON LIVE CHAINS IN A BROWSER

Minutes of RPC calls become *seconds.*

Polkadot Asset Hub, newest 80,000 blocks

114,460 rows · 23,268 accounts · loaded in Pantry



Portaldot V3 history, blocks #1–#129,069

snapshot + direct-synced tail · 938 / 938 account checks reconciled



1.8 s

to verify one account: 183 rows (L2) + its balance (L3)

133 / 133

Kusama AH recovery timelines replay to on-chain storage

2,173,842

rows from 1.5 M Polkadot AH blocks, 30 / 30 chunks verified

Every number comes from real chains; reproduce with the CLI and Pantry

PORTALDOT V3 TESTNET · NODE-CONSOLE.PORTALDOT.DEV

Deployed, published, *verifiable.*

ITEM	ADDRESS · RANGE · TRANSACTION	STATUS
PotluckRegistry (EVM)	0xef9305561eb0c34d205dfa0d9099a105f44282eb block #129,014 · tx 0x3abb9058e1eb48628791a029de71ef11b510ebc7a5cfa81677425773ce0bc352	LIVE
Manifest 2 · Portaldot history	blocks #1–#129,069 · stored at #129,092–#129,095	PUBLISHED
Manifest 3 · Polkadot AH	#22,021,128–#22,101,127 · 114,460 rows · 3.2 MB · 372.64 tPOT	PUBLISHED
Manifest 4 · Kusama AH recovery	#17,836,634–#22,802,789 · stored at #218,503–#218,507	PUBLISHED
PotluckRegistry (PVM)	resolc build passes ReviveApi.instantiate dry run · needs ≈ 3,293 tPOT	PENDING

All hashes: contracts/deployments/portaldot-v3-testnet.json

STATIC FILES, LIVE CHAINS, NO BACKEND

Two apps you can open *right now.*

Pantry

wngstnr-code.github.io/potluck/pantry/

- Catalog of published histories, read from the registry
- Portaldot from genesis by direct sync: 2,248 rows in 2 min 25 s, then L3 905 / 905 in 22.7 s
- Polkadot AH from its snapshot in 5.6–6.4 s, with per-account L2 + L3
- Attack demo: watch L3 catch a forged chunk

5.6–6.4 s
Polkadot AH, 80,000 blocks

905 / 905
Portaldot accounts reconciled

The SDK's showroom: browse, load and verify any history.

Wasiat-lite

wngstnr-code.github.io/potluck/wasiat/

- Kusama AH Recovery: 121 plans, 14 open attempts, 13 inheritors
- Per-account timelines: who the friends are, who started, who cancelled as proof of life
- Set up the same friend group on Portaldot, signed in your wallet (fee ≈ 0.0223 tPOT)

121
inheritance plans on Kusama AH

133 / 133
timelines replay to storage

An inheritance app must outlive its maker's servers. Potluck makes that possible.

EVERY POLKADOT-SDK DAPP SHOWS HISTORY

The same data, without *the server or the trust.*

APPROACH	SERVERS TO RUN	WHY USERS BELIEVE IT	SURVIVES ITS MAKER	BIG-CHAIN LOAD
Hosted indexer API	The provider's	**Trust the API**	**No**	Fast
Self-hosted indexer	**Yours, always on**	**Trust your database**	**While you pay**	Fast
Direct RPC sync	**None**	**Reads the chain**	**Yes**	**Days**
Potluck	**None**	**Checked against the chain**	**Yes, data on Portaldot**	**Seconds**

Wallets and explorers

Account history on Polkadot and Kusama Asset Hub

Long-lived apps

Recovery, inheritance, governance records that must outlast a team

Portaldot builders

A free unified two-VM index for every Hacker House team

BUSINESS MODEL AND ECOSYSTEM VALUE

Every stored byte *is paid in POT.*

01 **Protocol fee**
A small fee on top of storage when publishing or renewing through the official SDK.

02 **Index on demand**
dApps post bounties to keep a spec fresh; publishers are paid per checkpoint that passes verification.

03 **Potluck Studio**
Local-first spec builder, simulator and manifest monitor, with a paid team tier.

04 **Chain partnerships**
Data-service subsidies from Portaldot and from polkadot-sdk chains without an indexer.

≈ 100 POT / MB
stored on Portaldot (testnet rate), paid by publishers; readers pay nothing

0.6–1 GB
gzip of Polkadot AH balance history over its Revive window alone: one chain, one spec

Renewals
every blob renewed each ~7 days: recurring on-chain activity, not a one-off upload

First users: Hacker House teams, Asset Hub developers, Kusama Recovery users

ROADMAP

Working today. *Ready to grow.*

DONE	Registry, publisher CLI, browser SDK, Pantry and Wasiat-lite Three chains published as manifests 2–4 on Portaldot 1.5 M Polkadot AH blocks built and verified, 30 / 30 chunks
NEXT	Deploy the PVM registry (≈ 3,293 tPOT) Publish the 1.5 M-block build (≈ 6,200 tPOT)
AFTER	Backfill to Revive activation (block #11,405,259) Index-on-demand bounties and Potluck Studio More polkadot-sdk chains as sources

Portaldot becomes the history warehouse for the Polkadot ecosystem.

Built solo by @wngstnr-code during Hacker House 2026: 150+ commits since Oct 20, tests on recorded real chain data, 21 contract tests at 100% coverage.

`repo` `github.com/wngstnr-code/potluck`
`site` `wngstnr-code.github.io/potluck`
`registry` `0xef93…82eb`